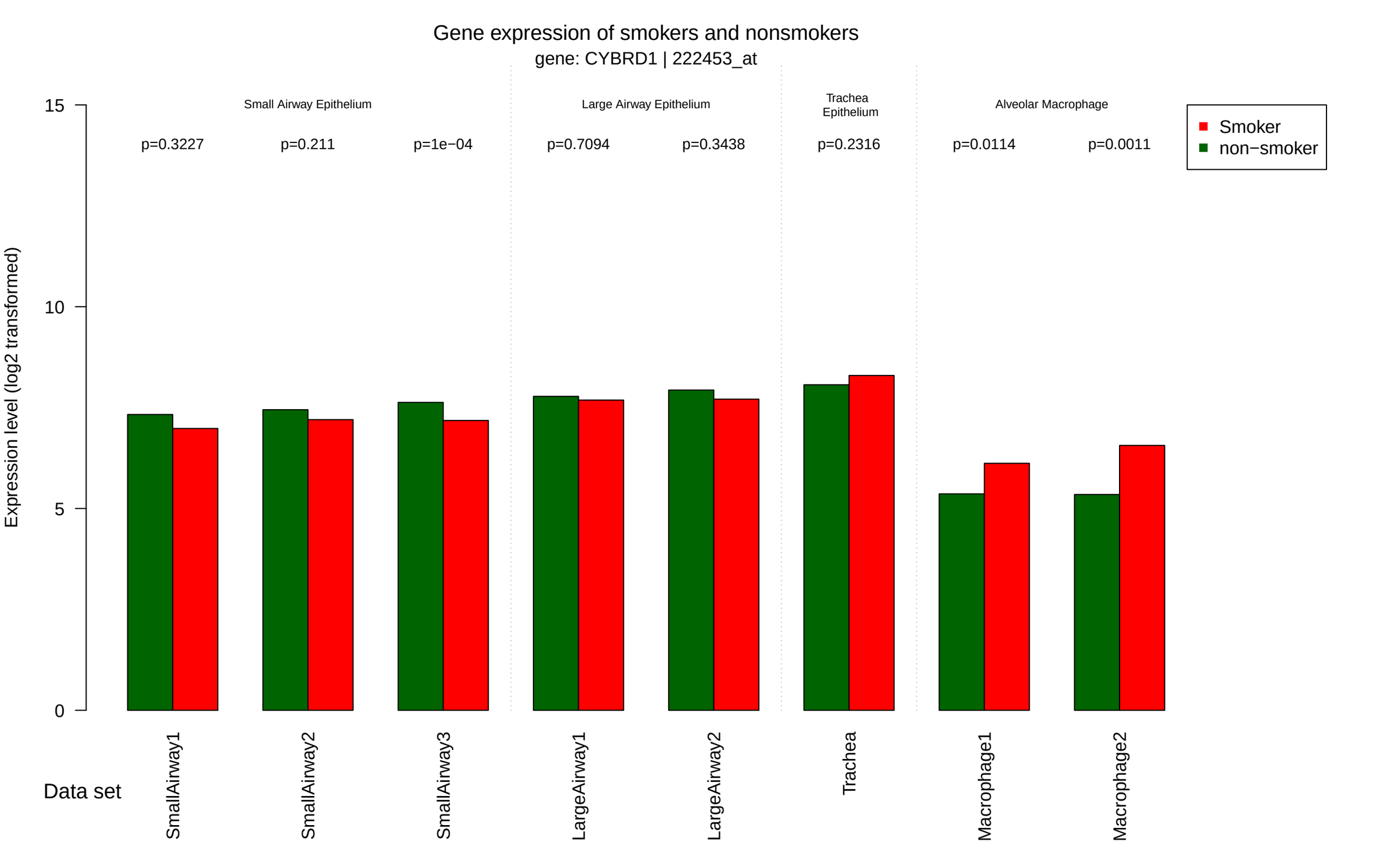
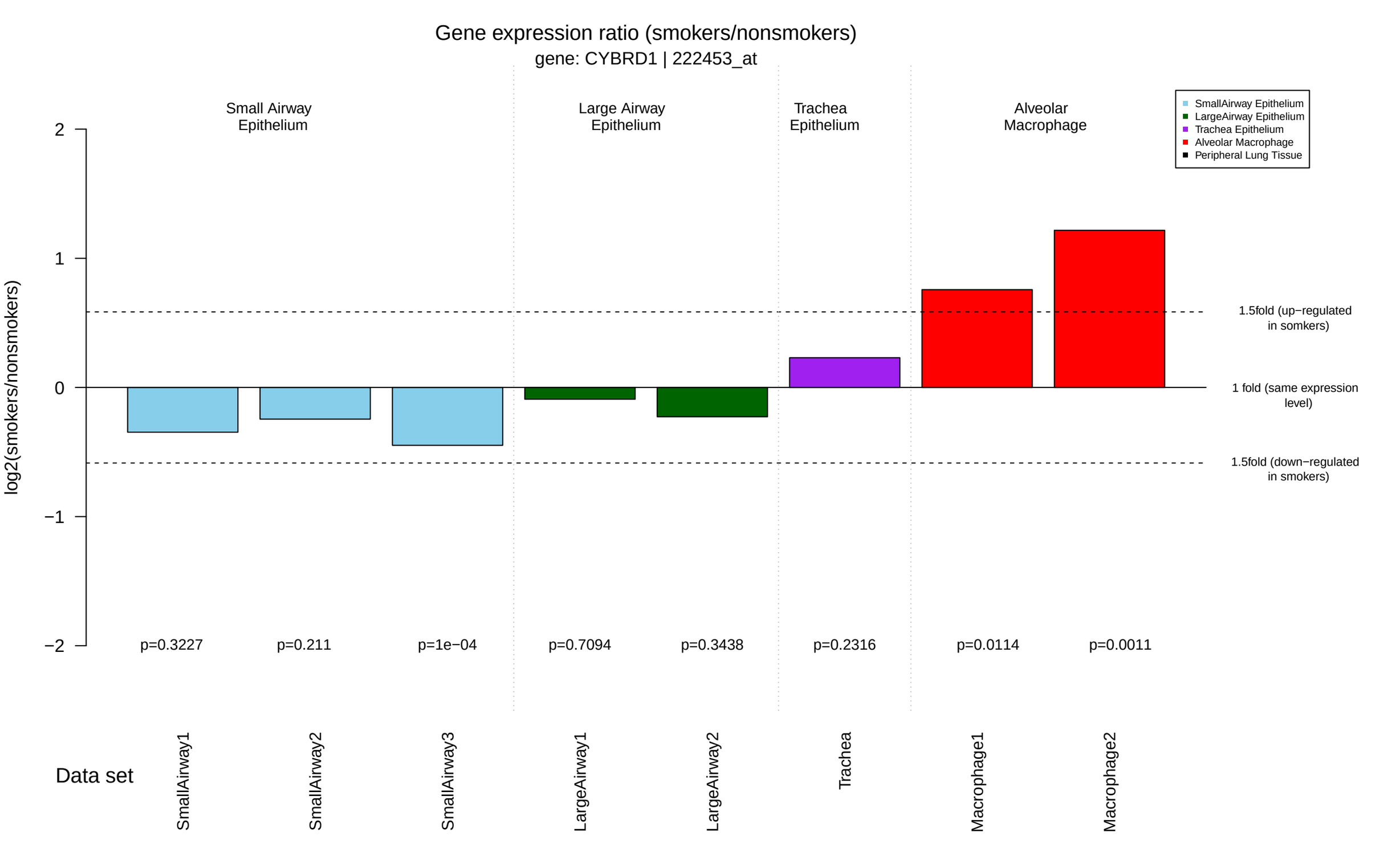
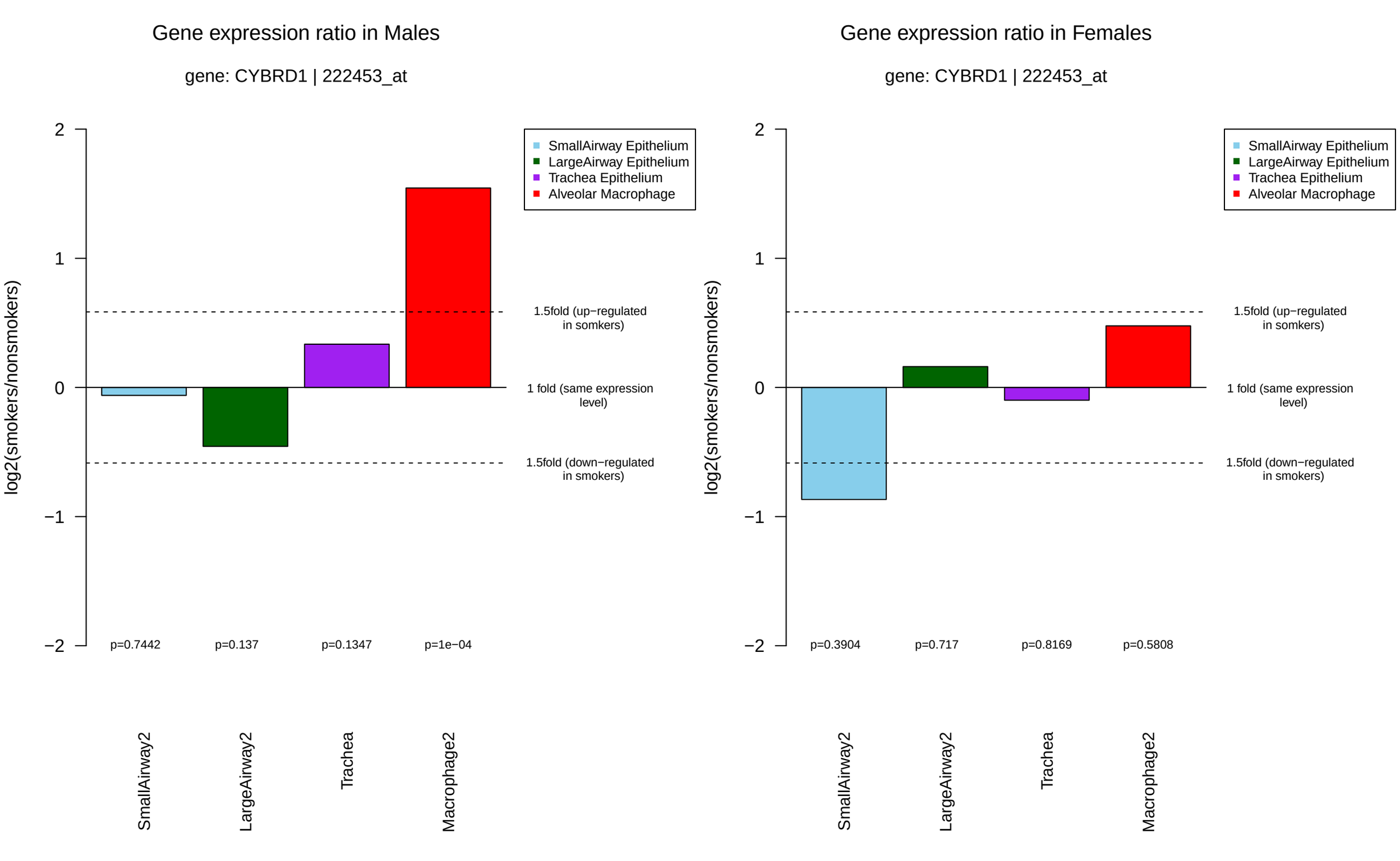
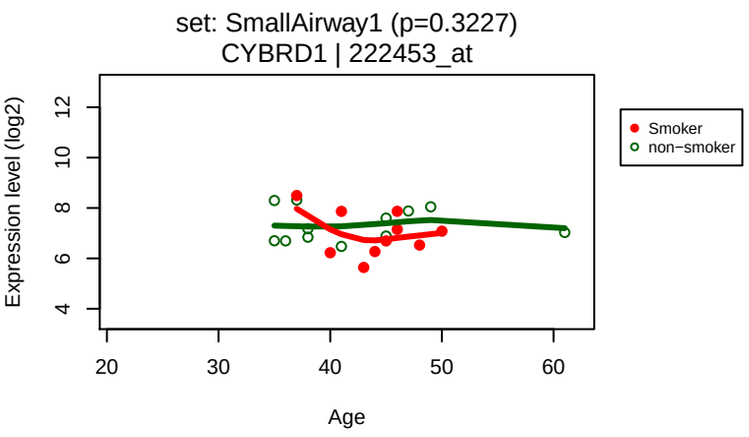
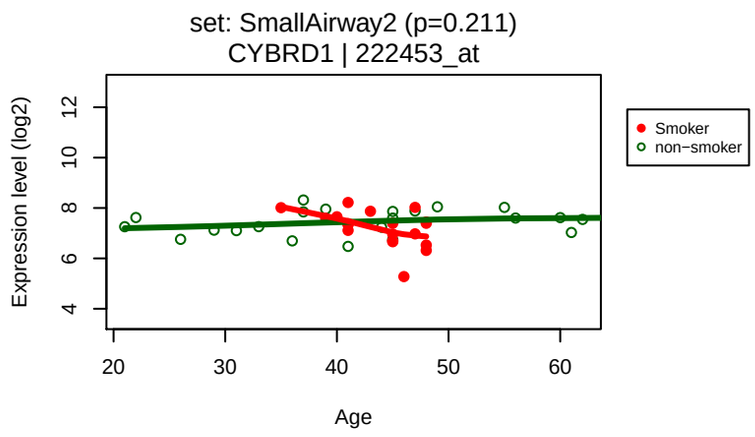
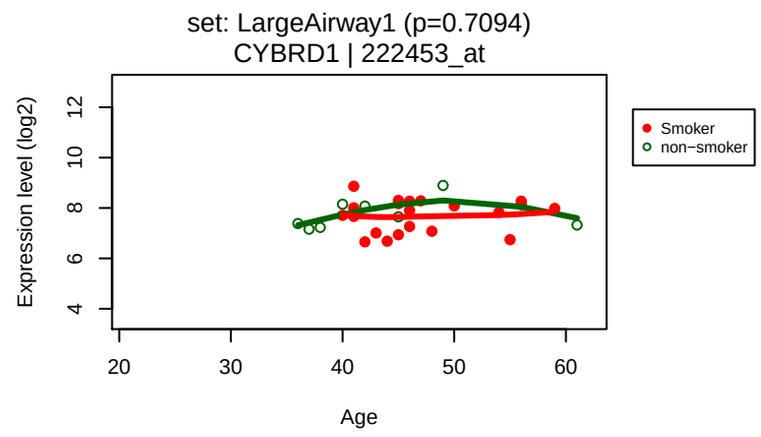

Gene expression of smokers and nonsmokers
gene: CYBRD1 | 222453_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3227
p=0.211
p=1e−04
p=0.7094
p=0.3438
p=0.2316
p=0.0114
p=0.0011
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

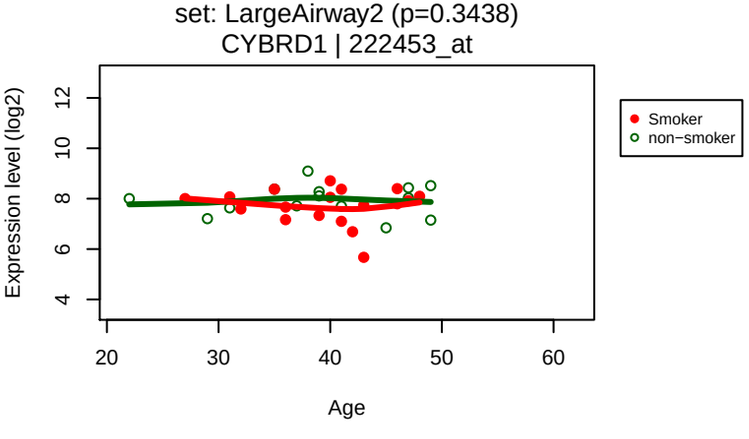

Gene expression ratio (smokers/nonsmokers)
gene: CYBRD1 | 222453_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3227
p=0.211
p=1e−04
p=0.7094
p=0.3438
p=0.2316
p=0.0114
p=0.0011
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

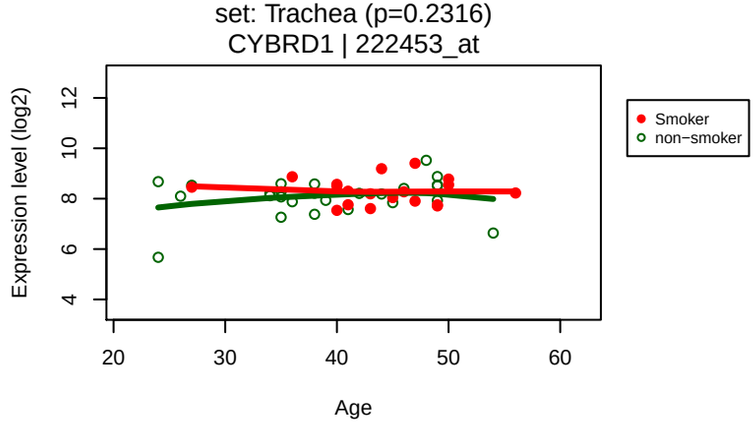

Gene expression ratio in Males
gene: CYBRD1 | 222453_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7442
p=0.137
p=0.1347
p=1e−04
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CYBRD1 | 222453_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3904
p=0.717
p=0.8169
p=0.5808
SmallAirway2
LargeAirway2
Trachea
Macrophage2

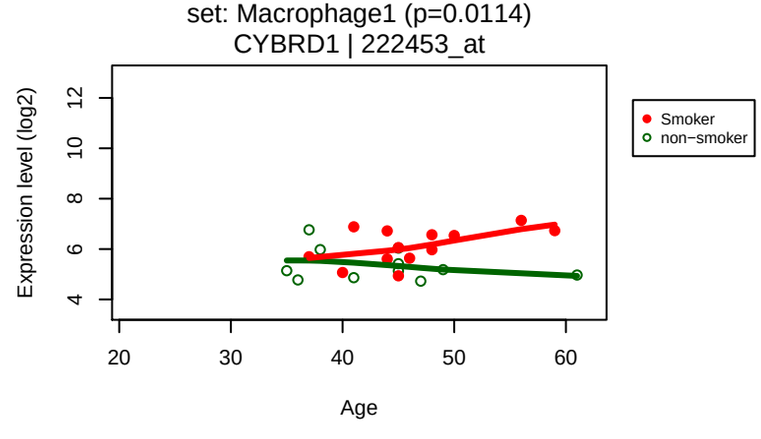

set: SmallAirway1 (p=0.3227)
CYBRD1 | 222453_at
Expression level (log2)
Age
Smoker
non−smoker

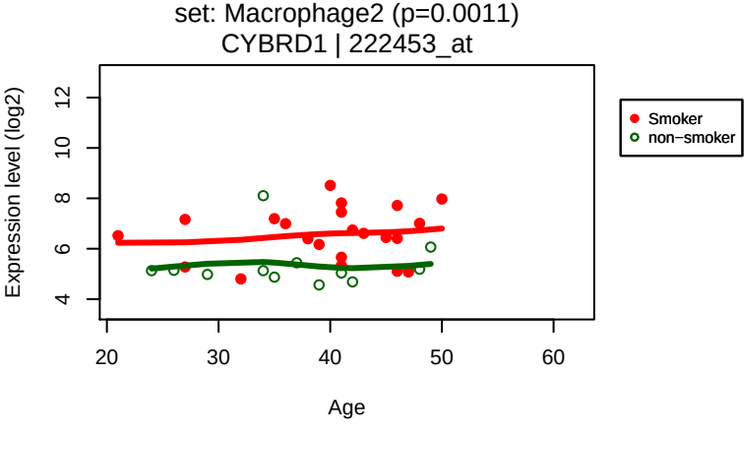

set: SmallAirway2 (p=0.211)
CYBRD1 | 222453_at
Expression level (log2)
Age
Smoker
non−smoker

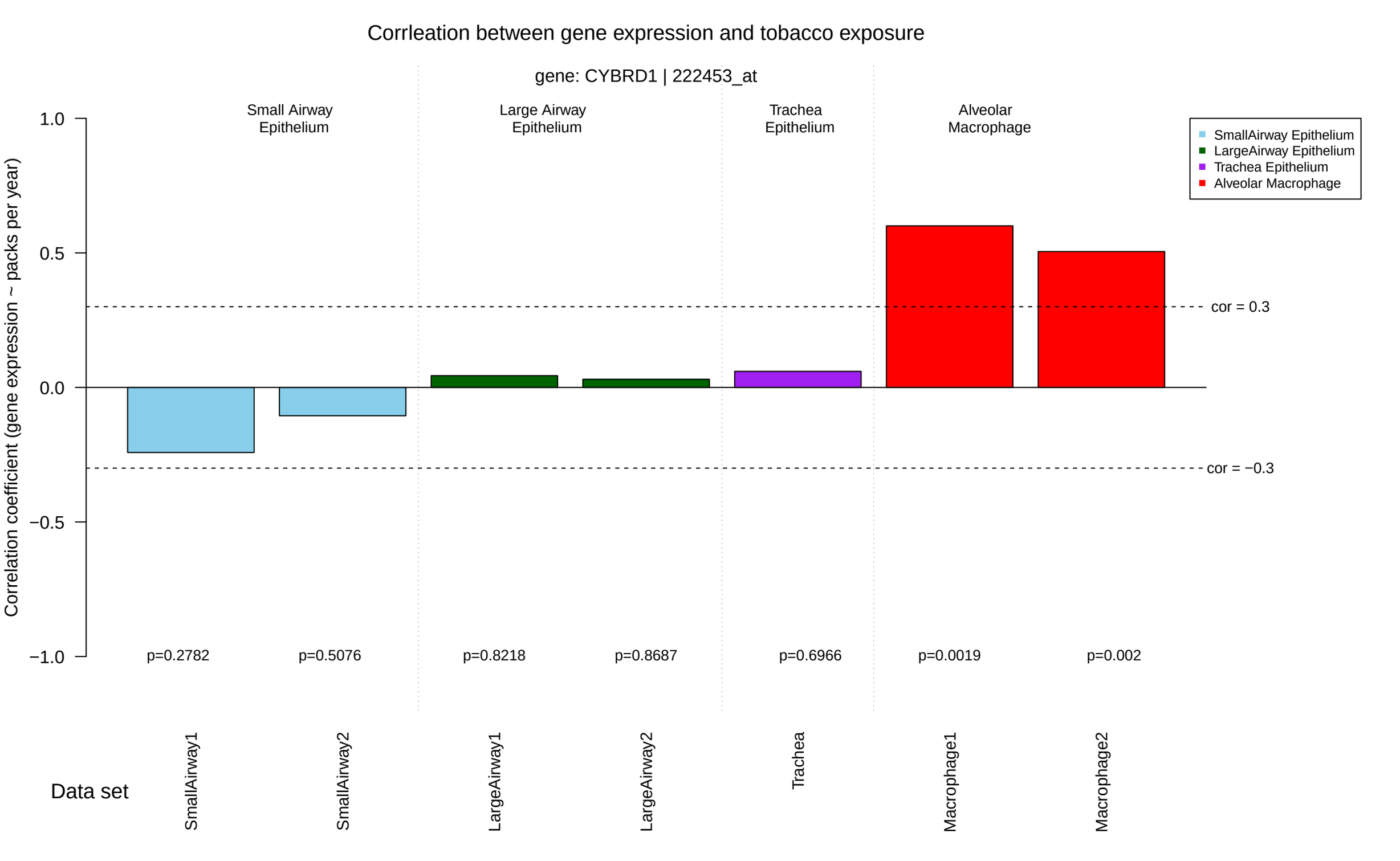

set: LargeAirway1 (p=0.7094)
CYBRD1 | 222453_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3438)
CYBRD1 | 222453_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2316)
CYBRD1 | 222453_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0114)
CYBRD1 | 222453_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0011)
CYBRD1 | 222453_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CYBRD1 | 222453_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2782
p=0.5076
p=0.8218
p=0.8687
p=0.6966
p=0.0019
p=0.002
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2